Sum of Four Squares

Time limit: 1.00 s
Memory limit: 512 MB

CSES 3355 | Mathematics | https://cses.fi/problemset/task/3355

A well known result in number theory is that every non-negative integer can be represented as the sum of four squares of non-negative integers.

You are given a non-negative integer n. Your task is to find four non-negative integers a, b, c and d such that $n = a^2 + b^2 + c^2 + d^2$.

Input

The first line has an integer t: the number of test cases.

Each of the next t lines has an integer n.

Output

For each test case, print four non-negative integers a, b, c and d that satisfy $n = a^2 + b^2 + c^2 + d^2$.

Constraints

- $1 \le t \le 1000$
- $0 \le n \le 10^7$
- the sum of all n is at most 10^7

Example

Input

```
3
5
30
322266
```

Output

```
2 1 0 0
1 2 3 4
314 159 265 358
```